

Monad Mini PC Node System Report

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Generated: 2026-05-22 22:55:31 MDT

Privacy posture: public report. Hostnames, usernames, IP addresses, keys, signatures, wallet-like hex strings, and token-like values are redacted or omitted.

Executive Summary

Monad services: monad-bft=active, monad-execution=active, monad-rpc=active.

Estimated Monad service CPU over the prior interval: approximately 491.2% of one CPU core.

Thermals: max 89.2 C, average 42.0 C.

PDF resource-window screenshot: included (captured window: System Monitor).

System Specs

CPU: AMD Ryzen 7 8745H w/ Radeon 780M Graphics

CPU count reported by kernel: 16

Online CPU range: 0-7

RAM: 44Gi

Root disk: 915G total, 201G used, 669G available, 24% used

Kernel: 6.17.0-1023-oem

Monad Node Runtime

Services:

- monad-bft: active=active, state=running, main_pid=34003, restarts=0, memory=3535.2 MB, cpu_since_start=544.1 sec

- monad-execution: active=active, state=running, main_pid=33969, restarts=0, memory=11783.0 MB, cpu_since_start=70408.6 sec

- monad-rpc: active=active, state=running, main_pid=33959, restarts=0, memory=283.0 MB, cpu_since_start=1567.1 sec

Processes: no visible Monad process rows were returned by ps.

Temperature And Fluctuation Analysis

- Current thermal snapshot: average 42.0 C, maximum 89.2 C across 11 reported sensors.

- Four-hour comparison: 41.2 C increase in max temperature from the prior report.

- Thermal fluctuation note: the max temperature changed by +41.2 C while estimated Monad service CPU usage was 491.2%. Recent service events were present.

- 24-hour observed max-temperature range: 48.0 C to 89.2 C.

Top reported sensor readings:

- k10temp-pci-00c3 / Tctl: 89.2 C

- nvme-pci-0100 / temp1: 57.5 C

- amdgpu-pci-c500 / edge: 57.0 C

- spd5118-i2c-0-50 / temp1: 40.8 C

- spd5118-i2c-0-51 / temp1: 35.5 C

- nvme-pci-0400 / Sensor 2: 34.9 C

- nvme-pci-0400 / Composite: 31.9 C

- nvme-pci-0400 / Sensor 1: 31.9 C

- nvme-pci-0100 / Composite: 31.9 C

- nvme-pci-0100 / Sensor 1: 31.9 C

- acpitz-acpi-0 / temp1: 20.0 C

System And Node Events Cross-Check

Recent sanitized events from the last four hours, classified for public context:

- [Report collection issue] monad-bft: local report probes could not collect a non-critical

data point.

Sanitized detail: probe failed: Command '['journalctl', '-u', 'monad-bft', '--since', '4 hours ago', '--no-pager']' timed out after 20 seconds

- [Node-impacting] monad-execution: service lifecycle or resource events that can affect node availability.

Sanitized detail: May 22 19:38:43 [redacted-host] systemd[1]: Stopped monad-execution.service

- "Service file for Monad Execution".

- [Informational warning] monad-execution: warning-level message retained for context, with no clear local node impact.

Sanitized detail: May 22 19:38:43 [redacted-host] systemd[1]: Started monad-execution.service

- "Service file for Monad Execution".

- [Informational warning] monad-execution: warning-level message retained for context, with no clear local node impact.

Sanitized detail: May 22 19:38:43 [redacted-host] monad[33969]: 2026-05-23 01:38:43.735716286 [33969] event.cpp:321 LOG_WARNING file system hosting event ring file '/[redacted-token]-2MB/event-rings/monad-exec-events' does not support MAP_HUGETLB!

- [Node-impacting] monad-rpc: service lifecycle or resource events that can affect node availability.

Sanitized detail: May 22 19:38:43 [redacted-host] systemd[1]: Stopped monad-rpc.service -

"Service file for Monad RPC".

- [Informational warning] monad-rpc: warning-level message retained for context, with no clear local node impact.

Sanitized detail: May 22 19:38:43 [redacted-host] systemd[1]: Started monad-rpc.service -

"Service file for Monad RPC".

- [Informational warning] monad-rpc: warning-level message retained for context, with no clear local node impact.

Sanitized detail: May 22 19:39:43 [redacted-host] monad-rpc[33959]:

{"timestamp":"2026-05-23T01:39:43.495450Z","level":"ERROR","fields":{"message":"Error traversing db while populating triedb cache txs","block_key":"Proposed(ProposedBlockKey(76379520, 62f...

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{"timestamp":"2026-05-23T01:39:43.495507Z","level":"ERROR","fields":{"message":"Error traversing db while populating triedb cache receipts","block_key":"Proposed(ProposedBlockKey(76379520...

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{"timestamp":"2026-05-23T01:39:43.495519Z","level":"ERROR","fields":{"message":"Error traversing db while populating triedb cache txs","block_key":"Proposed(ProposedBlockKey(76379520, 62f...

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Security Change Log

Recent operator security changes, sanitized for public sharing:

- 2026-05-07: Restored Monad runtime settings with systemd drop-ins instead of editing vendor unit files.

Why: Keeps the node configuration easier to audit and preserve across package updates while restoring the known-good event-ring, RPC websocket, and CPU-placement behavior from the earlier install.

Privacy note: No command lines, passwords, keys, peer addresses, node identities, wallet-like values, or public IP addresses are recorded here.

- 2026-05-07: Restricted Monad secret-bearing files to owner-only read/write permissions.

Why: The local keystore environment file and node identity key files were more broadly readable than necessary. Owner-only permissions reduce local accidental disclosure risk without changing node behavior.

Privacy note: File paths are summarized by purpose; key contents and unlock values are never included.

- 2026-05-07: Confirmed normal local users cannot read the Monad BFT process command line or environment through procfs.

Why: The BFT binary currently accepts its keystore unlock value through a command-line option, so process-argument visibility matters. Existing procfs hiding reduces exposure to non-privileged local users.

Privacy note: The actual process command line and keystore unlock value are intentionally omitted.

- 2026-05-07: Documented the remaining upstream security limitation for Monad BFT keystore unlock handling.

Why: A cleaner design would let operators pass the keystore unlock value through a protected file, stdin, systemd credential, or similar non-argv mechanism. The current local binary help only exposes a command-line unlock option.

Privacy note: This notes the design limitation only; no secret [redacted] is recorded.

- 2026-05-07: Kept public report generation sanitized and avoided raw systemctl status or full process command-line capture.

Why: Full status and process listings can expose sensitive command-line arguments. Reports use narrower systemd fields, sanitized journal excerpts, and explicit redaction before writing TXT/PDF outputs.

Privacy note: Reports redact hostnames, usernames, IP addresses, keys, signatures, wallet-like hex strings, token-like values, and sensitive argument values.

- 2026-05-10: Added a local OpenTelemetry Collector for the Monad metrics endpoint.

Why: Monad BFT requires a telemetry endpoint argument. The previous configuration pointed at a local endpoint with no listener, causing repeated connection-refused errors. A localhost-only collector now provides the required endpoint without exporting private telemetry externally.

Privacy note: The report records only the operational change and reason. No raw telemetry payloads, peer identities, public IP addresses, keys, command lines, or unlock values are included.

Research Notes

- This report is generated from local system telemetry and Monad service logs on the mini PC.

- Values are snapshots, not lab-grade benchmarks. Trends become more useful after multiple four-hour samples.

- The report intentionally avoids private identity material, validator keys, wallet-like values, raw peer lists, public IPs, and command lines containing sensitive arguments.

- Intended use: public community reporting about running a Monad node on smaller hardware while retaining useful operational data for technical readers.

System Resource Window Snapshot

